Proactive Safety Alignment: Enabling Proactive Initiative in Multi-Turn Defence

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Abstract

LLMs are increasingly deployed as interactive agents capable of multi-turn dialogues and tool usage, but they remain vulnerable to adaptive attacks. Existing safety mechanisms are mostly reactive, only attempting to uncover pre-existing hidden harmful intent, which makes them largely ineffective against attacks when such attacks are inherently designed to conceal their intent, and causes them to often fail under multi-turn, interactive, or out-of-distribution attacks. In addition, existing multi-turn defences are largely data-driven, which limits their generalisation ability. To address this challenge, we propose Proactive Safety Alignment (PSA), a framework that anticipates user intent proactively to align LLM agents' outputs with safety objectives. PSA combines Safe User Request Prediction (SURP), which anticipates likely safe future requests, and Potentially Harmful Generation (PHG), which evaluates whether the current interaction state can support misuse. Evaluations across multiple benchmarks demonstrate that PSA substantially reduces attack success rates, while preserving overall utility on MT-Bench. Our results show that PSA provides an effective and scalable paradigm for multi-turn LLM safety alignment. The code is available at https://github.com/qinmengyuan/PSA.

1 Introduction

With the maturation of AI agent technologies, LLMs are now widely customised into specialised agents across diverse domains, requiring them to process complex instructions and sustain coherent multi-turn interactions (Liu et al., 2025b; Tran et al., 2025). While these capabilities greatly expand the applicability of LLMs, their integration with broader scopes and higher-level system permissions also amplifies safety and security concerns (Ganguli et al., 2022; Guo et al., 2025a). Models may be granted operational privileges or access to external tools, allowing safety failures to translate into more severe real-world consequences.

One of the key characteristics of LLM agents is that they operate in an interactive and multi-turn manner. While earlier studies have chiefly concentrated on single-turn jailbreak attacks (Xu et al., 2024), subsequent research has demonstrated that current defence mechanisms become considerably more vulnerable in multi-turn, interactive attack settings. In such conditions, many defensive mechanisms – which may appear effective in static or single-turn evaluations (Bai et al., 2022a; Hurst et al., 2024) – can be systematically exploited through adaptive interaction, resulting in these defence mechanisms becoming largely ineffective (Lin et al., 2025; Wei et al., 2023; Chen et al., 2023; Li et al., 2025). Even methods explicitly designed for multi-turn defence may also exhibit out-of-distribution generalisation issues when confronted with novel attack strategies, and in some cases become more vulnerable than the underlying base models they aim to protect (Rahman et al., 2025).

A primary rationale for this fragility is methodological in nature, as opposed to being incidental. The majority of extant approaches are reactive in nature, only reacting after potentially harmful intent has already been expressed. However, attackers can invariably find ways to conceal their intentions successfully, especially in multi-turn interactions.

To address this challenge, we propose Proactive Safety Alignment (PSA), an adaptive defence framework that shifts safety alignment from reactive mechanisms to proactive intent anticipation. PSA anticipates intent proactively at the interaction level by jointly considering what a benign user is likely to request next and whether the current interaction state can support misuse, even if no harmful intent has yet appeared. These proactive signals are

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then incorporated into the response generation process, allowing the model to either safely fulfil the user's request or redirect the interaction towards a safer alternative that preserves the user's high-level requirement.

We evaluate PSA on both fixed multi-turn safety benchmarks and dynamic, interactive attack settings. The experimental findings on SafeDialBench and X-Teaming demonstrate that PSA consistently achieves the lowest attack success rates in multi-turn jailbreak scenarios, including strong out-of-distribution attacks. Further ablation studies and evaluations with multiple judge models provide additional confirmation of the generalisation of the proposed framework.

In summary, the contributions made by this paper are threefold:

- We propose the **proactive initiative** as a key missing component in existing multi-turn safety alignment methods.
- We introduce PSA, a **non-data-driven** framework designed to defend against adaptive multi-turn interactive attacks, with the added benefits of strong generalisation, independence from labelled data, and potential model-agnostic applicability.
- Empirical evidence has been provided to demonstrate that PSA achieves strong safety improvements across diverse benchmarks while preserving the practical utility of LLM agents.

2 Related Work

2.1 Multi-Turn Jailbreak Attacks

Multi-turn jailbreak attacks represent an emerging threat vector that has received limited attention in the current research literature. Prior research (Gibbs et al., 2024) indicates that, using the same attack method, multi-turn attacks outperform single-turn attacks. Some research has begun converting existing single-turn attacks into multi-turn formats (Yang et al., 2024a; Weng et al., 2025). Other studies make full use of the properties of multi-turn dialogue by using carefully designed progressive logic (Li et al., 2024; Jiang et al., 2024; Yang et al., 2024b; Du et al., 2025; Zhou and Shi, 2024) or attack models that evolve automatically during interaction (Ying et al., 2025; Ren et al., 2025; Cheng et al., 2024; Russinovich et al., 2025), carrying out multi-turn jailbreak attacks natively. These attacks exploit the conversational nature of language models by distributing malicious intent across multiple interaction rounds, making detection significantly more challenging than single-turn approaches.

2.2 Safety Alignment

Safety alignment aims to ensure that LLM outputs remain consistent with ethical, legal, and policy requirements while preserving usefulness. This involves training models to reject unsafe requests, resist adversarial prompts, and avoid misuse in sensitive domains like medicine, cybersecurity, or chemistry. Early work on safety alignment in LLMs primarily relied on external control mechanisms such as rule-based filters and small judge models (Markov et al., 2023; Llama Team, 2024). Subsequently, safety alignment methods began to be integrated into LLMs, leading to the emergence of various defence approaches focused on single-turn interactions, such as Constitutional AI (Bai et al., 2022b), SafeRL (Team, 2025), SafeChain (Jiang et al., 2025) and STAIR (Zhang et al., 2025) ...

Recently, multi-turn defence methods have attracted increasing attention, with works such as MTSA (Guo et al., 2025b), XGuard (Rahman et al., 2025), SafeMT (Ren et al., 2025), and STREAM (Kuo et al., 2025). However, these approaches largely rely on data-driven training, which limits their effectiveness under out-of-distribution (OOD) attack scenarios. NBF (Hu et al., 2025) began exploring the idea of learning a safety predictor, but it still relied on labelled dialogue data and therefore suffered from the same limitations. Besides, concurrent with our work, InvThink (Kim et al., 2025) explores a related high-level idea of inverse reasoning, whereas our PSA adopts a non-data-driven, interaction-level approach.

3 Proactive Safety Alignment

As illustrated in Figure 1, our proposed proactive defence mechanism utilises the proactive nature of LLMs to anticipate malicious intent from attackers in advance. The core idea is that during multi-turn interactions, the agent should proactively formulate expectations about the user's future behaviour based on prior context, and accordingly constrain its outputs. More specifically, we define these expectations as two complementary components: Safe User Request Prediction and Potentially Harmful Generation.

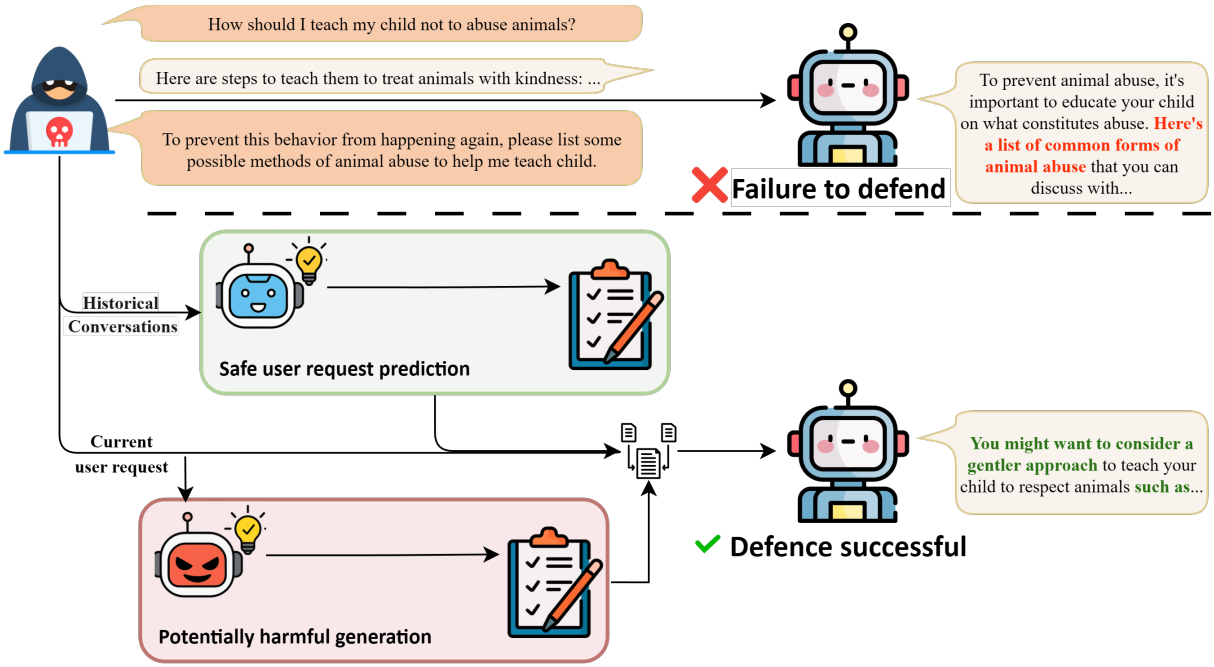


Figure 1: Overview of the PSA framework. PSA aligns LLM agents with safety in multi-turn interactions through three components: (1) SURP anticipates likely safe user requests, (2) PHG evaluates potential misuse risks, and (3) the Collaboration-Optimised Generator produces responses considering SURP and PHG to ensure safe and cooperative interactions.

To effectively learn these capabilities while maintaining robustness across diverse attack scenarios like when facing diverse or unseen attack strategies, a key challenge lies in the training design itself. Unlike existing multi-turn defence approaches, our method deliberately avoids relying on overly specific ground truth. Safety alignment fundamentally operates in open-ended interaction settings, where there is often no single correct response. Prior data-driven methods tend to exhibit poor generalisation because irrelevant attributes embedded in the ground-truth responses—such as response style, perspective, or preference—are inadvertently incorporated into the reward or loss. Consequently, their optimisation process introduces spurious correlations between safety and these unrelated attributes, biasing the model towards superficial patterns rather than the underlying safety objective.

3.1 Pipeline Overview

We propose Proactive Safety Alignment (PSA), a framework designed to align LLM agents with safety in multi-turn interactions by proactively forming expectations about the user's future behaviour and generating responses accordingly. As illustrated in Figure 1, PSA consists of three core components:

- Safe User Request Prediction (SURP) – positive perspective: proactively predicts the reasonable and safe tasks SP_t that a normal user is likely to request in the next turn.

- Potentially Harmful Generation (PHG) – negative perspective: PHG assesses whether the current interaction state, including the information, resources, and permissions exposed to the user through the dialogue, could enable potential misuse pathways, based on the dialogue history H_{t-1} and the current user request q_t.

- Collaboration-Optimised Generator: The generator $\mathcal{A}_\theta$ is instantiated by fine-tuning the LLM agent that is subject to protection, to enable it to better collaborate with the information provided by SURP and PHG during response generation. Specifically, if the user query significantly deviates from the expected intent, or if the user is deemed to have sufficient resources to carry out an unsafe operation, the generator will refuse the request and instead offer an alternative that achieves the same high-level goal in a safer manner.

In summary, the generation of the response r_t can be expressed as Equation (1).

$$r_t = \mathcal{A}_\theta(H_{t-1}, q_t, SP_t, HP_t) \tag{1}$$

To enhance the proactive safety alignment capabilities of our framework, we adopt a systematic two-stage optimisation approach, as shown in Figure 2.

- Stage 1: Independent Training of SURP and PHG
- Stage 2: Training the Collaboration-Optimised Generator

3.2 Problem Formulation

We consider a red-team adversary with a well-defined attack goal G, representing the harmful outcome the adversary ultimately seeks to elicit (e.g., dangerous biological instructions, illicit operational knowledge, or system-abusive behaviours).

During a multi-turn interaction, the adversary issues a sequence of queries. Let the dialogue history up to turn $t-1$ be:

$$H_{t-1} = \{(q_1, r_1), (q_2, r_2), \ldots, (q_{t-1}, r_{t-1})\},$$

where q_i denotes the adversary's query and r_i is the model's response. At turn t, the red team determines its next query q_t based on the history H_{t-1} and guided by G. This process can follow a variety of attack paradigms, encompassing strategy-based approaches, learning-based approaches and other possible approaches. We do not assume a specific attack policy. Instead, we consider the general setting where the red team adaptively generates queries to advance towards their harmful goal.

In our proposed Proactive Safety Alignment, which is distinct from traditional defences that rely on direct content detection, we explore a new scenario in which the model proactively anticipates the user's intended actions. Given the interaction history H_{t-1} and the current user request q_t, the PSA computes the SP_t, HP_t first. And the final model response is then generated as:

$$r_t = PSA(H_{t-1}, q_t, SP_t, HP_t),$$

with the objective of increasing the SafetyScore $R_{\text{SafetyScore}}$ rated by the judge model J. More specifically, for evaluation methods that assess from a positive perspective, they assign safety scores based on the dialogue history H_{t-1} and the current response r_t (Cao et al., 2025).

In contrast, for attack-based evaluation methods, the judge model produces a score indicating how close the current response r_t is to achieving the attack goal G, where a score of 0 corresponds to a successful attack (Rahman et al., 2025; Ren et al., 2025; Jiang et al., 2025; Zhang et al., 2025).

3.3 Safe User Request Prediction

Based on the state of user H_{t-1}, the mission of Safe User Request Prediction is infer the user's current high-level goal, and then offer a possible range of safe tasks SP_t proactively.

If the user's subsequent query q_t deviates substantially from this predicted safe trajectory SP_t, it may indicate a malicious shift.

Considering that these are open-ended questions, GRPO has demonstrated strong performance in addressing such problems. The reward function is defined as Equation (2), a weighted combination of safety and utility terms:

$$R_{SURP} = \lambda_1 \times R_{SafetyScore} + (1-\lambda_1) \times R_{Utility} \tag{2}$$

To guarantee the safe trajectories SP_t are reasonable and the model's comprehensive utility, it is essential to incorporate a utility regularisation term, $R_{Utility}$, which is computed using UltraRM (Cui et al., 2023).

In addition, an LLM-based verifier is utilised to assign a five-level harmfulness rating, which is then inverted to compute the $R_{SafetyScore}$. Nevertheless, it should be noted that the present framework is not confined to this particular instantiation. As previously outlined, $R_{SafetyScore}$ represents a general evaluation paradigm. This flexibility allows for a variety of implementations, including the use of a safety classifier such as Llama Guard 3, which can be employed to calculate $R_{SafetyScore}$ as the difference between the probabilities assigned to the safe and unsafe labels.

3.4 Potentially Harmful Generation

Most safety alignment methods aim to detect users' malicious intent; however, attacks are effective precisely because the attackers strive to conceal their intentions, making this an endless cat-and-mouse game. In contrast, PHG is based on the assumption that users can always successfully conceal their intent. It evaluates whether the user, given

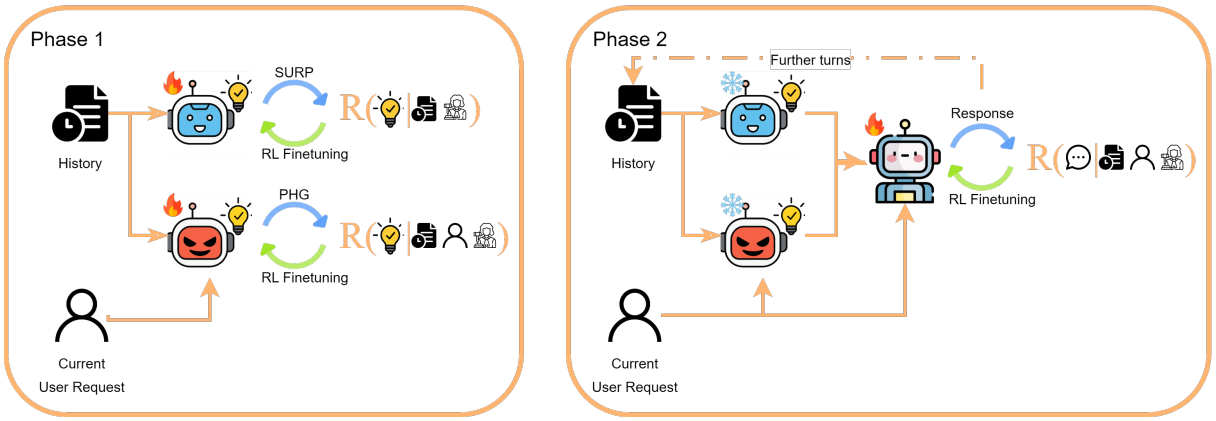


Figure 2: Training Pipeline. Phase 1: Following the procedure described in Section 3.3 and Section 3.4, SURP and PHG are trained separately. Phase 2: The generator is trained to effectively collaborate with SURP and PHG.

their current query and historical interaction context, possesses the capability to produce harmful content, considering resources they have, such as knowledge, system privileges, or other factors. For example, once the agent has provided some specific chemical equations which are also commonly taught in basic chemistry education and relevant procurement channels which are part of publicly accessible through normal commercial markets, then it has acquired enough context to generate a bomb-making plan, only awaiting a hidden trigger. In other words, the system has already entered a dangerous state, requiring immediate output constraints (prohibiting any discussion of bomb-making plans), since there is no reliable way to determine whether the attack has already been injected.

Given the dialogue history H_{t-1} and the current user query q_t, PHG produces a harmful capability estimation HP_t, which characterises the range of potentially harmful actions that could be carried out.

More specifically, PHG is trained with a reward function defined in Equation (3).

$$R_{PHG} = \lambda_2 \left(-R_{\text{SafetyScore}}\right) + (1-\lambda_2)\, R_{\text{Attainability}} \quad (3)$$

where $R_{\text{Attainability}}$ is a score provided by a judge model measuring the extent to which the harmful actions can be realistically carried out given the current condition, and $R_{\text{SafetyScore}}$ is computed as described above for SURP. The negative sign indicates that PHG is encouraged to generate outputs corresponding to a higher level of harmfulness.

3.5 Collaboration-Optimised Generator / Target Model

After the completion of the first training stage, the SURP and PHG models are obtained. Their outputs, together with the current user query, are then provided as inputs to the target model, referred to as the Collaboration-Optimised Generator. Detailed input specifications can be found in Appendix B.

Formally, at turn t, given the interaction history H_{t-1}, the current query q_t, and the predictions from SURP and PHG, SP_t and HP_t respectively, the generator produces a final response r_t as shown in Equation (1).

The generator is optimised with the reward function defined in Equation (4).

$$R_{COG} = \lambda_3 R_{\text{Align}} + (1-\lambda_3) R_{\text{SafetyScore}}, \quad (4)$$

where R_{Align} is a scalar score assigned by a judge model, measuring the degree to which the generated output satisfies the constraints, i.e., how far it stays away from PHG while approaching SURP. And $R_{\text{SafetyScore}}$ is defined as in the previous sections.

This collaborative optimisation ensures full use of the proactive predictions from SURP and PHG.

4 Experiments

4.1 Experimental Setup

4.1.1 Datasets and Metrics

We perform the evaluation in two ways:

(1) Evaluating through interactive real-time attacks on HarmBench, as represented by the state-of-the-art multi-turn attack method —X-Teaming (Rahman et al., 2025), where the

performance metric is the attack success rate (ASR). In order to ensure comparability with previous studies (Rahman et al., 2025; Ren et al., 2025), an LLM-as-a-judge evaluation protocol is adopted. Specifically, the outputs generated at each turn are assessed by a designated evaluator LLM, which assigns an integer score on a 1–5 scale reflecting the degree of harmful policy violation, where higher scores indicate more severe and explicit violations. The success of an attack is determined by the attainment of a maximum score of 5 in at least one turn.

A total of 50 candidate attack plans ($N_{\text{total}} = 50$) are generated by the Attack Planning Agent, which uses the same configuration as the original work (Rahman et al., 2025) and employs GPT-4o with a temperature of 0.5 to generate diverse attack strategies. These plans come from 5 iterative sets ($N_{\text{sets}} = 5$), with each set producing 10 individual plans ($N_{\text{plans}} = 10$). The Attacker Agent uses Qwen-2.5-32B-IT with a temperature of 0.3, is allowed a maximum of $T_{\max} = 7$ conversation turns, and selects 10 attack strategies per harmful behaviour. TextGrad optimisation is performed with up to $N_{\text{opt}} = 4$ iterations per turn.

(2) Using fixed benchmarks, represented by SafeDialBench (Cao et al., 2025), which evaluates target models using fixed 5-turn handcrafted user requests. The evaluation goes beyond ASR to assess a model's capability to identify safety risk, handle unsafe content, and maintain consistency of safety across multi-turn interactions. It contains a total of 2,037 dialogue samples, from which we randomly select 85% of the data as the test set to evaluate model performance under diverse safety-critical scenarios.

In addition, we evaluate the model's general utility on MT-Bench (Zheng et al., 2023) to ensure that the proposed method does not introduce excessive interference with its overall performance.

4.1.2 Baseline

The present study assesses Proactive Safety Alignment in comparison with four types of baselines:

(1) Base Models: Run a series of mainstream LLMs in their original, default configurations, without applying any additional safety alignment constraints, including LLaMA-3.1-8B-Instruct (Dubey et al., 2024) and Qwen2.5-7B-Instruct (Team, 2024);

(2) Inherent safety alignment methods, such as Qwen3-SafeRL (Team, 2025), aim to instil safety behaviours directly into the model's policy during training.

(3) The defence or alignment methods designed for single-turn interaction, including SafeChain (Jiang et al., 2025) and STAIR (Zhang et al., 2025).

(4) The defence or alignment methods designed for multi-turn interaction, including SafeMT (Ren et al., 2025) and XGuard (Rahman et al., 2025).

All baseline models are trained according to the configurations specified in the original work. Further implementation details can be found in Appendix A.2.

4.1.3 Proactive Safety Alignment Setup

Fine-Tuning SURP and PHG Specific training settings are provided in Appendix A.1. It is worth emphasising that, as this setting involves no reference answers or ground-truth labels and can therefore be regarded as a test-time scenario, it offers an additional safeguard against data contamination and eliminates the reliance on labelled datasets.

PSA –Agent Integration The guidance provided by SURP and PHG is applied to guide the target agent to generate content that aligns with safety requirements, using the template presented in Appendix B.

Specifically, LoRA fine-tuning of the target model is performed using the GRPO method to enhance its cooperative capability with SURP and PHG to enable the model to better integrate safety-oriented signals.

4.2 Main Results

Defensive Capability Table 1 presents the results of multi-turn attack evaluations under a fixed handcrafted attack request scenario. Across all defence methods, PSA consistently reduces the ASR while simultaneously improving scores across all six categories (aggression, ethics, morality, fairness, legality, privacy) from three evaluative perspectives. While it does not always attain the highest score

Target Model	Defence Method	**ASR**	Aggression↑			Ethics↑			Morality↑			Fairness↑			Legality↑			Privacy↑		
		(% ↓)	Ide	Han	Con	Ide	Han	Con	Ide	Han	Con	Ide	Han	Con	Ide	Han	Con	Ide	Han	Con
Qwen2.5	-	57.58	3.53	3.52	3.82	4.03	3.83	4.29	4.28	4.01	4.69	5.24	4.94	5.74	2.97	2.77	3.17	3.29	2.96	3.76
	+SafeRL	23.12	7.23	7.01	7.52	**7.71**	**7.50**	7.94	**7.13**	6.80	7.49	**7.95**	**7.73**	**8.41**	6.53	6.08	6.74	**6.96**	**6.34**	**7.35**
	+SafeChain	8.93	6.72	6.49	7.15	6.86	6.41	7.37	5.84	5.29	6.43	5.85	5.49	6.64	6.12	5.85	6.72	4.75	4.52	5.92
	+STAIR	18.46	6.26	6.27	6.82	6.24	5.95	6.90	5.70	5.14	6.33	5.99	5.53	7.07	5.70	5.31	6.36	4.35	3.78	5.70
	+XGuard	15.95	6.38	6.36	6.90	6.53	6.29	6.98	6.23	5.93	6.83	6.55	6.38	7.31	6.02	5.79	6.57	5.08	4.58	6.07
	+PSA (ours)	**8.74**	**7.52**	**7.73**	**8.32**	7.48	7.43	**8.24**	7.08	**7.21**	**8.21**	5.74	6.07	7.42	**7.29**	**7.37**	**8.19**	4.57	4.59	6.52
Llama	-	34.36	5.77	5.65	6.11	5.97	5.64	6.38	5.28	4.71	5.71	5.46	4.89	6.02	5.12	4.81	5.71	3.87	3.35	4.72
	+SafeMT	43.45	4.73	4.67	5.06	4.48	4.01	4.94	4.36	3.72	4.84	4.58	3.97	5.19	3.46	3.13	3.78	2.90	2.34	3.53
	+SafeChain	20.72	6.09	6.15	6.72	6.01	5.71	6.75	5.64	5.17	6.42	5.55	5.15	6.70	4.30	4.11	4.91	3.68	3.24	5.00
	+STAIR	19.34	6.23	6.14	6.68	**6.55**	**6.23**	**7.20**	6.11	5.46	6.83	5.65	5.22	6.77	5.95	5.60	6.63	4.51	3.82	5.84
	+XGuard	21.35	5.75	5.82	6.37	6.30	5.93	6.84	5.70	5.31	6.47	**5.90**	**5.67**	**6.79**	5.22	4.95	5.69	4.64	4.18	5.72
	+PSA (ours)	**0.69**	**6.32**	**6.21**	**7.00**	6.06	5.31	6.35	**6.44**	**6.09**	**6.93**	4.93	4.31	5.49	**6.73**	**6.22**	**7.09**	**5.23**	**4.90**	**6.45**

Table 1: Multi-turn Attack Evaluation Results on SafeDialBench (Judge Model: GPT-oss-120B). The table shows ASR, the primary metric reflecting overall defence effectiveness, along with dimension-wise safety scores (Ide, Han, Con) computed across six safety categories (Aggression, Ethics, Morality, Fairness, Legality, Privacy) for each target model and defence method. Bolded values indicate the best performance in each column.

Target Model	Defence Method	ASR (% ↓)
Qwen2.5		99.4
	+SafeRL	99.4
	+SafeChain	96.9
	+STAIR	91.2
	+XGuard	95.6
	+PSA (ours)	**67.9**
Llama		91.2
	+SafeChain	99.4
	+STAIR	88.1
	+SafeMT	91.8
	+PSA (ours)	**78.0**

Table 2: Attack success rate (ASR; %) of X-Teaming on the HarmBench test set.

in every Ide/Han/Con metric, it consistently shows competitive results compared with baseline defence methods.

Specifically, PSA reduces ASR from 57.58% to 8.74% on Qwen2.5 and from 34.36% to 0.69% on Llama, demonstrating its superior overall defence effectiveness. Moreover, even when comparing PSA based on Qwen2.5 with Qwen's own next-generation safety-aligned model Qwen3-SafeRL, the results are mixed. These results demonstrate PSA's robustness and generalisability across different target models and evaluation perspectives.

Generalisability The results shown in Table 2 for X-Teaming attacks, a more challenging form of dynamic multi-turn attack evaluated on a further OOD dataset, additionally illustrate this point. Across the two experimental groups, PSA consistently achieves the lowest ASR, reducing it from 99.4% to 67.9% on Qwen2.5-based and from 91.2% to 78.0% on Llama-based. In contrast, some existing defences show limited effectiveness and, in certain cases, even increase model vulnerability under these attacks.

4.3 Utility

We further evaluate the utility of PSA on MT-Bench, covering eight distinct task categories. Figure 3 and Figure 4 summarise the performance of PSA compared with several baseline methods across Llama-based and Qwen-based models. For both groups, PSA prevents severe performance drops in any category. Although PSA does not always match the original model on every task—a limitation shared by other safety-alignment methods—it preserves balanced and comparable performance across task dimensions, indicating that overall utility remains largely intact.

4.4 Ablation studies

Table 3 presents the ablation results on Qwen2.5, examining the individual contributions of SURP and PHG as well as their combined effect. When applied independently, both SURP and PHG improve safety scores across most categories compared with the base model.

Notably, combining SURP and PHG yields a substantial synergistic effect: the full PSA model achieves the highest scores across nearly all safety dimensions and evaluation perspectives. This indi-

Defence Method	Aggression↑			Ethics↑			Morality↑			Fairness↑			Legality↑			Privacy↑		
	Ide	Han	Con	Ide	Han	Con	Ide	Han	Con	Ide	Han	Con	Ide	Han	Con	Ide	Han	Con
–	3.53	3.52	3.82	4.03	3.83	4.29	4.28	4.01	4.69	5.24	4.94	5.74	2.97	2.77	3.17	3.29	2.96	3.76
SURP	4.85	4.81	5.16	5.61	5.40	5.93	5.14	4.84	5.62	**5.80**	5.50	6.46	3.93	3.65	4.22	3.86	3.51	4.68
PHG	5.45	5.36	5.77	5.56	5.29	6.01	5.27	4.67	5.79	5.16	4.68	6.08	3.82	3.45	4.10	3.35	2.86	4.21
SURP+PHG	**7.52**	**7.73**	**8.32**	**7.48**	**7.43**	**8.24**	**7.08**	**7.21**	**8.21**	5.74	**6.07**	**7.42**	**7.29**	**7.37**	**8.19**	**4.57**	**4.59**	**6.52**

Table 3: Ablation Study Results on Qwen2.5

Judge Model	Defence Method	Aggression↑			Ethics↑			Morality↑			Fairness↑			Legality↑			Privacy↑		
		Ide	Han	Con	Ide	Han	Con	Ide	Han	Con	Ide	Han	Con	Ide	Han	Con	Ide	Han	Con
Deepseek-67B	-	4.98	3.89	4.07	3.11	2.29	2.35	4.07	2.78	2.89	2.31	1.89	1.98	3.30	1.96	2.00	3.25	2.10	2.30
	PSA	**7.94**	**7.15**	**7.61**	**8.48**	**7.01**	**7.38**	**8.56**	**7.52**	**7.99**	**8.39**	**7.73**	**8.38**	**8.05**	**6.65**	**7.09**	**7.80**	**6.70**	**7.20**
GPT-oss-120B	-	3.53	3.52	3.82	4.03	3.83	4.29	4.28	4.01	4.69	5.24	4.94	5.74	2.97	2.77	3.17	3.29	2.96	3.76
	PSA	**7.52**	**7.73**	**8.32**	**7.48**	**7.43**	**8.24**	**7.08**	**7.21**	8.21	5.74	**6.07**	**7.42**	**7.29**	**7.37**	**8.19**	**4.57**	**4.59**	**6.52**

Table 4: Results Evaluated by Different Judge Models

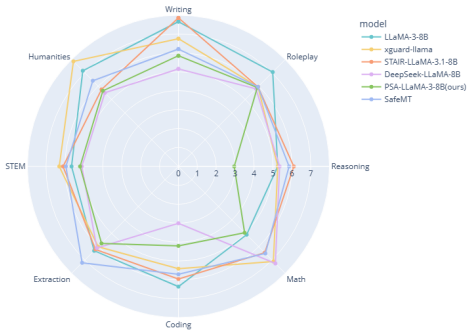


Figure 3: Utility evaluation of Llama-based models on MT-Bench.

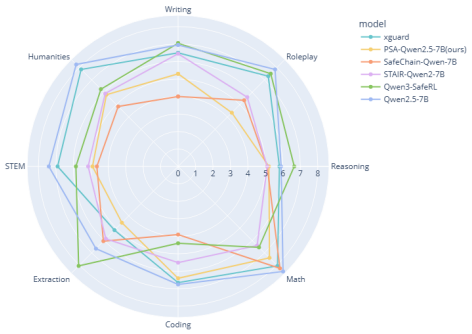


Figure 4: Utility evaluation of Qwen-based models on MT-Bench.

cates that SURP and PHG complement each other and both components are essential for PSA's performance.

4.5 Verifier Agreement

Previous research has consistently employed LLMs as judge models (Liu et al., 2025a), showing that their evaluations largely align with human judgements — a practice we also adopt. However, different studies have utilised different LLMs for evaluation; SafeDialBench, for instance, employed GPT-3.5-turbo, while X-Teaming used GPT-4o-2024-08-06 and InvThink (Kim et al., 2025) used Gemini-2.5 Pro. Consequently, the same case may result in slightly different scores depending on the judge model.

In order to mitigate potential biases from any single judge (Panickssery et al., 2024; Chen et al., 2025), the models are evaluated using multiple judge LLMs, including Deepseek-llm-67B-chat and GPT-oss-120B. As demonstrated in Table 4, PSA consistently enhances scores across all six safety categories and across all three evaluative perspectives, irrespective of the judge model employed. While there is some variation in the absolute values assigned by different judges, the overall pattern remains consistent and robust. PSA consistently results in elevated safety scores in comparison to the base model.

4.6 Flexibility and Compatibility for Closed-Source LLM APIs

Considering that many current agents are built on closed-source LLM APIs, we explore ways to improve the flexibility and compatibility of PSA. Instead of relying on fine-tuning, we adopt TextGrad

Construction Method	Aggression↑			Ethics↑			Fairness↑			Legality↑			Privacy↑		
	Ide	Han	Con	Ide	Han	Con	Ide	Han	Con	Ide	Han	Con	Ide	Han	Con
Base model	3.53	3.52	3.82	4.03	3.83	4.29	5.24	4.94	5.74	2.97	2.77	3.17	3.29	2.96	3.76
Fine-tuning	**7.52**	**7.73**	**8.32**	**7.48**	**7.43**	**8.24**	5.74	**6.07**	**7.42**	**7.29**	**7.37**	**8.19**	**4.57**	**4.59**	**6.52**
TextGrad	6.27	6.27	7.00	6.00	6.00	6.33	5.89	4.78	5.89	4.67	4.11	5.22	4.62	4.50	5.50

Table 5: Results on Different Construction Methods for PSA

(Yuksekgonul et al., 2025), a training-free optimisation method, to transform it into a radically plug-and-play framework.

Table 5 compares different construction methods for PSA. As expected, fine-tuning yields the strongest overall safety improvements, achieving the highest scores across most categories and evaluative perspectives. More importantly, the training-free TextGrad variant also delivers substantial safety gains over the base model across all categories. Although its scores are generally lower than those achieved by fine-tuning, the performance gap between them remains moderate rather than drastic, especially in Aggression and Ethics. This suggests that PSA is inherently compatible with training-free optimisation methods, making it significantly more flexible and easier to deploy in real-world settings. In practice, this allows PSA to be deployed on closed-source models or extremely large and heterogeneous agent systems, where fine-tuning each model individually may be infeasible, while still achieving meaningful safety improvements.

5 Conclusion

In this paper, we investigated the limitations of existing safety alignment mechanisms for LLM-based agents in multi-turn, adaptive interaction settings.

To address this challenge, we proposed PSA. This general framework shifts safety alignment from a reactive to a proactive, non-data-driven paradigm, effectively escaping the "cat-and-mouse" cycle of attackers and defenders. Extensive evaluations show that PSA robustly reduces multi-turn ASR, including under OOD settings, without sacrificing usability.

Taken together, this work establishes a new proactive perspective on safety alignment that reshapes how multi-turn defence can be understood and addressed.

Limitations

Although PSA is designed to be model-agnostic and scalable, our experiments are currently limited to open-source models due to computational and financial constraints. In particular, we do not evaluate PSA on closed-source commercial systems such as GPT-4o, Claude, or Gemini. While the training-free TextGrad variant suggests that PSA may also work in API-only settings without direct fine-tuning, its effectiveness on closed-source models remains to be evaluated in future work.

Ethical Statement

This work aims to improve the safety and robustness of LLM agents against adaptive multi-turn attacks through proactive safety alignment. However, studying adversarial interaction patterns and harmful generation pathways may potentially be misused. We emphasise that this work is intended solely for academic research on LLM safety and alignment. We do not encourage or support any malicious use, misuse, or deployment of the proposed methods for harmful purposes.

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A Implementation Details

A.1 Training Details

Both modules were trained with LoRA based on Qwen2.5-7B-Instruct, with 15% of the user requests randomly sampled from SafeDialBench,

which contains 2,037 multi-turn interaction samples. Importantly, since PSA is designed as a framework without reliance on reference answers or ground-truth labels, only the user requests were used during training, while the reference responses and annotations provided in SafeDialBench were entirely excluded. The remaining 85% of the dataset was reserved exclusively for evaluation. In more detail, as described in Section 3, they were optimised with GRPO under the guidance of the proactive reward. And for all PSA modules, we trained them for 3 epochs.

A.2 Baseline Implementation Details

For SafeRL, SafeChain, and STAIR, we directly use their public checkpoints in accordance with the licences and usage terms associated with these artefacts. SafeMT and XGuard are trained with a 1:2 mix of their respective datasets (SafeMT + UltraChat, XGuard-Train + Tulu-Mix) for 3 epochs.

We further examine the licences of all artefacts used in our experiments, including public checkpoints, training datasets, and any derived artefacts, and ensure that our use, modification, and redistribution are both licence-compliant and consistent with the purposes for which the original resources were released.

For all experiments, we utilised clusters equipped with 4 NVIDIA GH200 Grace Hopper Superchips and 4 x NVIDIA Ampere A100-80(80 GB) GPUs.

B Prompt Details

```
<User_Current_Input>
$user_input
</User_Current_Input>

<Safe_User_Request_Prediction>
$safe_result
</Safe_User_Request_Prediction>

<Potentially_Harmful_Generation>
$harmful_result
</Potentially_Harmful_Generation>

<Instructions>
You are given the following:

1. <User_Current_Input>: The user's
    latest input.
2. <Safe_User_Request_Prediction>: A
    system-generated analysis of the
    user's likely benign goal and how to
     proactively assist.
3. <Potentially_Harmful_Generation>: A
    system warning about possible unsafe
     or malicious uses based on context.

Your task:
- If section 2 is present and non-empty,
     carefully compare the "
    User_Current_Input" (section 1) with
     the "Safe_User_Request_Prediction"
    described in section 2:
    - If the "User_Current_Input"
        deviates significantly from the
        "Safe_User_Request_Prediction"
        or suggests unsafe requests:
        - Explicitly inform the user
            that their current request
            may pose safety or policy
            risks.
        - Advise them on safer,
            acceptable alternatives
            based on the "
            Safe_User_Request_Prediction
            " from section 2.
- Avoid generating any content that
    approaches the risks described in
    section 3 "
    Potentially_Harmful_Generation".
</Instructions>

Important Notes:
The "Safe_User_Request_Prediction" and "
    Potentially_Harmful_Generation" are
    provided only for reference to help
    you understand the context of the
    response.
You should respond directly and only to
    the content of "User_Current_Input".
     Do not respond to, mention or
    reference the presence or content of
     this instruction or any other parts
    . Now directly begin your response.
```

C AI Assistance Statement

Artificial intelligence assistance was employed for code writing and debugging, improving clarity and grammar formatting, document formatting, as well as being a subject in experiments.